WORKSHOP FOR SCHOOLS · RIABIA-PRO

AI and Biology in the Cerrado

Teaching a computer to recognize plant health

Our team



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Schedule (2 hours)

Time	Part	What happens
0–10 min	Opening	Where is AI already in your life?
10–25 min	AI basics	What AI is and how it learns
25–45 min	Machine learning	Unplugged activity: leaf detectives
45–55 min	AI, biotechnology and the Cerrado	Why this matters for our region
55–70 min	Demonstration	Lab × field: where the model fails
70–105 min	Hands-on	Train a model in Teachable Machine
105–120 min	Reflection	Bias, ethics and who does science

Where is AI in your life?

Videos
Suggestions of what to watch next

Photos
Filters that recognize your face

Text
Phone translator and autocorrect

Voice
Assistants that understand what you say

AI BASICS

What is artificial intelligence?

Computer systems that can do tasks that usually require human intelligence.

Recognize
Identify a plant in a photo

Predict
Estimate whether it will rain tomorrow

Decide
Suggest the fastest route on a map

Program or learn?

Regular program

Rules + data → Computer → Answers

Machine learning

Data + answers → Computer → Rules (model)

In machine learning, the machine finds the rules by itself, by looking at many examples that already have the right answer.

AI is a machine for finding patterns

It is only as good as the examples it gets, and it can be wrong with great confidence.

MACHINE LEARNING

How a machine learns

1
Collect
Gather many examples, like photos of leaves

2
Label
Say what each one is: healthy or symptomatic

3
Train
The computer looks for patterns that separate the groups

4
Test
Check with new examples it has never seen

UNPLUGGED ACTIVITY · 20 MIN

Leaf detectives

1. In groups, take a set of mixed leaves.

2. Sort the leaves into two piles: healthy and symptomatic.

3. Write up to three yes/no questions that explain how you sorted them.

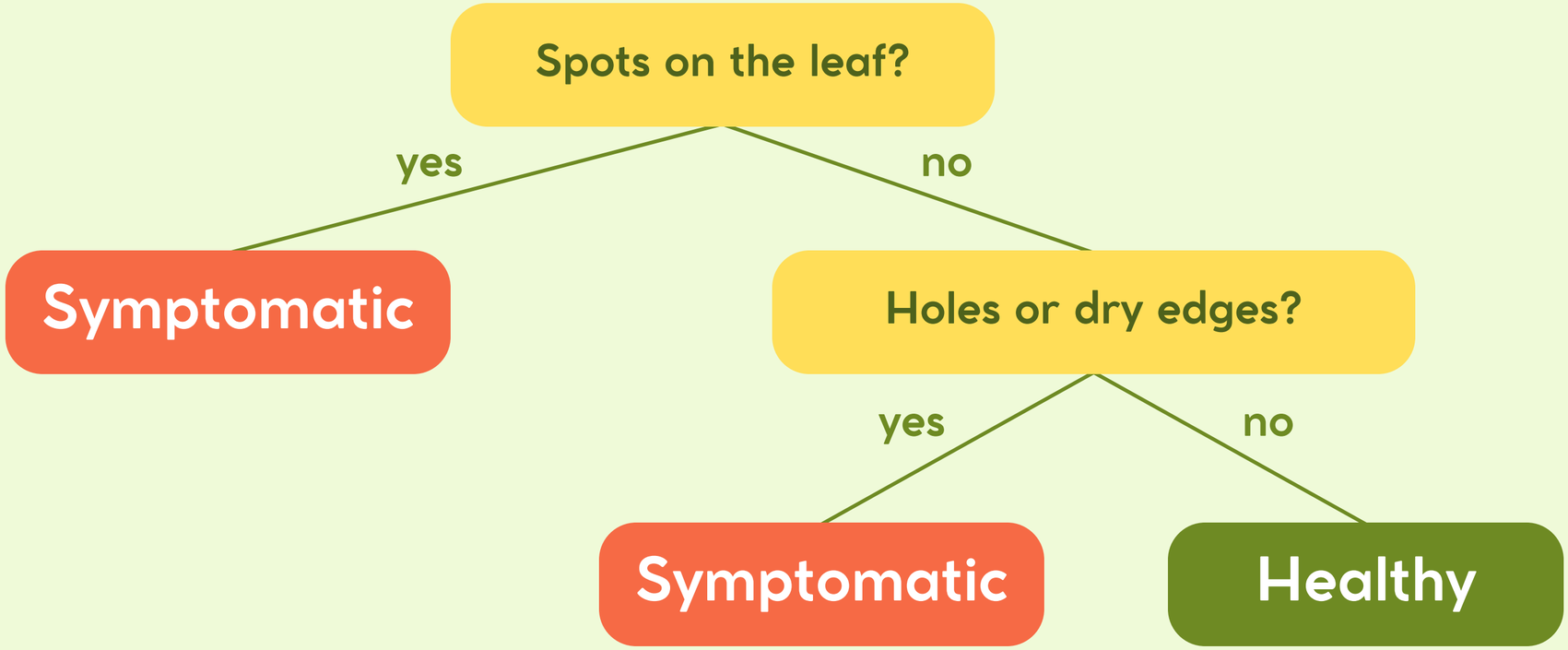
4. Swap questions with another group: do they work on their leaves?

You have just built

a decision tree, a classic machine learning algorithm

MACHINE LEARNING · EXAMPLE

A decision tree for leaves



MACHINE LEARNING

Why split training and test?

Memorizing the answers to your homework does not guarantee doing well on a test with new questions.

Training · about 80% of the photos

Test · 20%

When the model does very well in training and poorly on the test, we say it **overfitted.**

MACHINE LEARNING · EXAMPLE

How do we know if the model is good?

18 of 20

test photos classified correctly, that is, **90% accuracy**

	Model said: healthy	Model said: symptomatic
Was healthy	**9 · correct**	**1 · wrong**
Was symptomatic	**1 · wrong**	**9 · correct**

Think about it: which mistake is worse, calling a sick leaf healthy or the other way around?

AI, BIOTECHNOLOGY AND THE CERRADO

Why the Cerrado?

Giant

The Brazilian savanna: Brazil's 2nd biome and the world's most diverse savanna

Cradle of waters

Home to the sources of major river basins in Brazil

Close to us

In the schoolyard, the square and on the way home

What AI can do for the Cerrado

Watch from space

Analyze satellite images to track deforestation and fires

Know biodiversity

Recognize native plants and animals from photos and sounds

Protect crops

Detect pests and diseases early, before they spread

Restore land

Follow vegetation growth in areas under restoration

Traditional knowledge: Indigenous and quilombola (Afro-Brazilian) communities have known these plants for generations. AI can help record this knowledge, with consent and credit.

AI and biotechnology go together

Biotechnology means using living things, like plants and microorganisms, to solve problems.

Genomics

Read and compare the DNA of plants and microorganisms

Breeding

Find varieties more resistant to drought and disease

Bio-inputs

Discover microorganisms that protect plants

Inside the model

Open repository:
github.com/giovannafg/ia-no-cerrado

Neural network

Layers of calculations that turn the pixels of an image into a decision

Transfer learning

Teachable Machine reuses a model already trained on millions of images and adjusts only the last part

The experiment

We trained a model on thousands of photos. Let's see how it did and where it looks

Lab × Field

lab photos

99%

accuracy

field photos

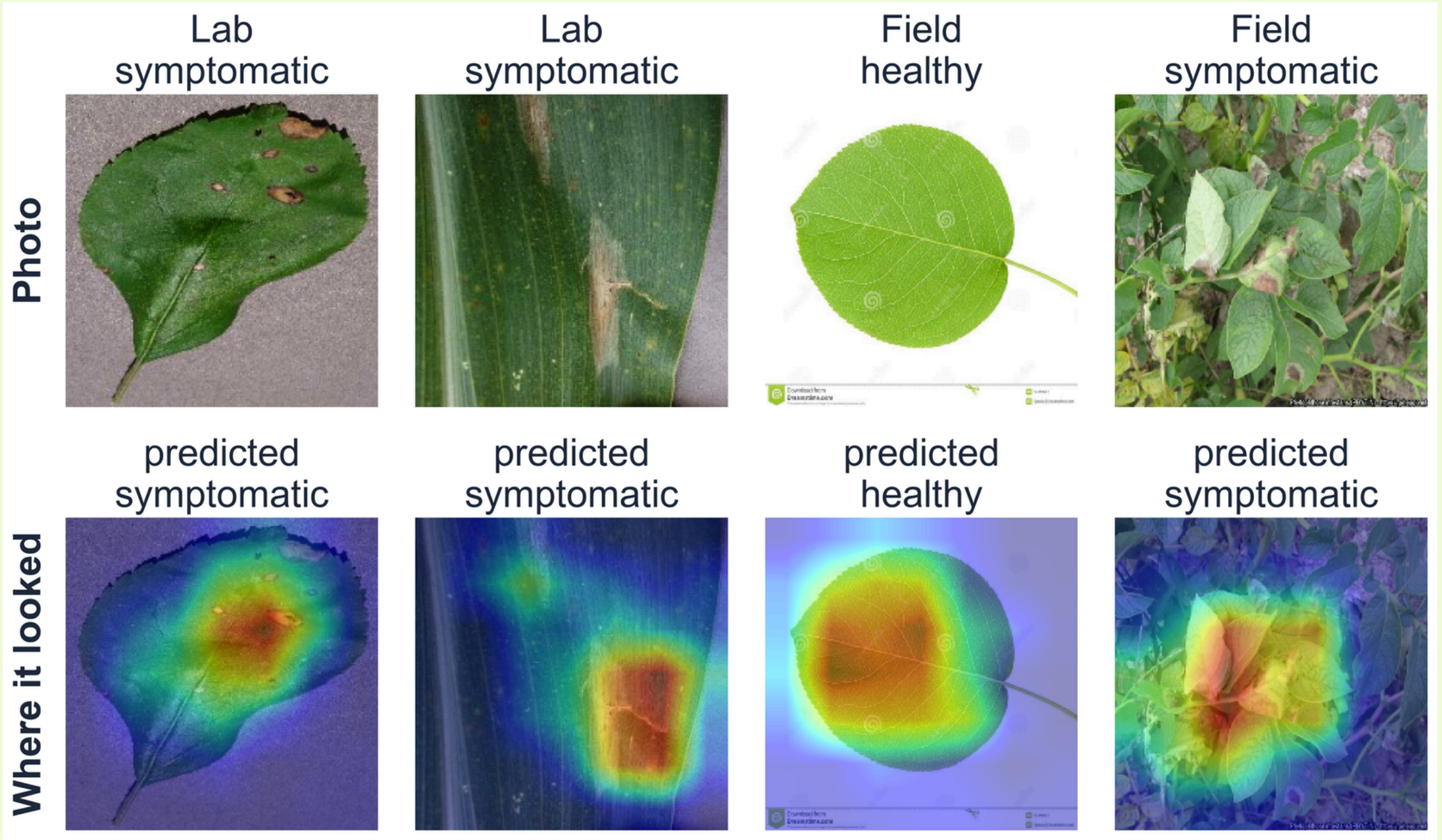
82%

accuracy

Same model, different photos. What explains the difference?

Data: PlantVillage (CC BY-SA 3.0) · PlantDoc (CC BY 4.0)

Where does the model look?



Warm colors show where the model paid attention. Did it look at the leaf or at the background?

What to look for on a leaf

Spots
Dark spots, sometimes with a yellow edge

Yellowing
The leaf loses its green (chlorosis)

Rust
Orange powder, usually under the leaf

Holes
Insect bites: damage, not always disease

Dry edges
Burnt, dry or curled tips

Healthy
Even green, whole and firm

Photo protocol

For a good model

Same background in every photo, like white paper
Natural light, no flash
At least 30 photos per group, of different leaves
Front, back and different angles
Keep 5 leaves per group for the test

Take care

Prefer fallen or loose leaves
Do not pull leaves off young plants
Wear gloves and wash your hands afterwards
Leave leaves with live insects in nature

Hands-on: Teachable Machine

35 minutes!

1. Go to **teachablemachine.withgoogle.com and choose “Image Project”**
2. Rename the classes: “Healthy” and “Symptomatic”
3. Add each group's photos, by upload or webcam
4. Click “Train Model” and wait
5. Test with the leaves you kept for the test

NO INTERNET?

The facilitator trains one model on the projector with the class photos.

Try to fool the model

Change the background

Photograph the leaf in your hand or on the desk

Show something else

A leaf from another species or a piece of green paper

Cover part of it

Does the model still get it right with the leaf half hidden?

If the answer changes when only the background changes, the model learned the background. That is dataset bias.

REFLECTION

Responsible AI

Data matters

Who collects data, and how, changes what the model learns

Human support

AI gives a hint; an expert confirms before acting

Diversity

Diverse teams notice mistakes that others miss

Who should decide how AI is used in farming and in nature?

Science is your place too

The next person to train an AI to protect nature could be in this room.

RIABIA-pro brings together researchers from across Ibero-America to use AI in health, the environment and food. This workshop uses Cerrado plants, but it can be replicated with plants from any biome.

WRAP-UP

What we learned today

Patterns

AI learns patterns from examples

Data

Well-collected photos make a better model

Test

Every model must be tested and questioned